Introdução ao Cálculo Diferencial e Integral

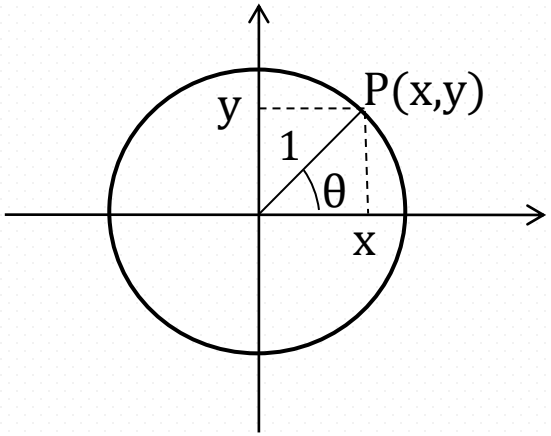
Funções Hiperbólicas

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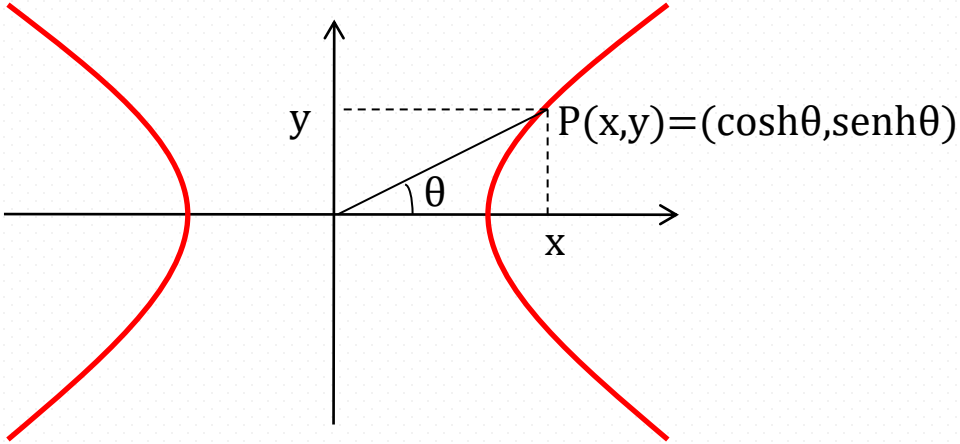
Funções Hiperbólicas

Das funções trigonométricas, temos que $P(x,y)=(\cos\theta,\operatorname{sen}\theta)$ está sobre uma circunferência de equação $x^2+y^2=1$. .



$$\left.\begin{aligned}\cos\theta &= \frac{x}{1} = x \\ \operatorname{sen}\theta &= \frac{y}{1} = y\end{aligned}\right\} \quad x^2 + y^2 = \operatorname{sen}^2\theta + \cos^2\theta = 1$$

Para as funções hiperbólicas, temos que $P(x,y)=(\cosh\theta,\operatorname{senh}\theta)$ está sobre uma hipérbole de equação $x^2-y^2=1$.



Funções Hiperbólicas

Definições:

$$\operatorname{senh}(x) = \frac{e^x - e^{-x}}{2} \quad \Rightarrow \text{Seno hiperbólico}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \quad \Rightarrow \text{Cosseno hiperbólico}$$

$$x^2 - y^2 = 1 \Rightarrow (\cosh(x))^2 - (\operatorname{senh}(x))^2 = 1$$

$$\Rightarrow \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = 1$$

$$\Rightarrow \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = \frac{4}{4} = 1$$

Funções Hiperbólicas

Gráficos

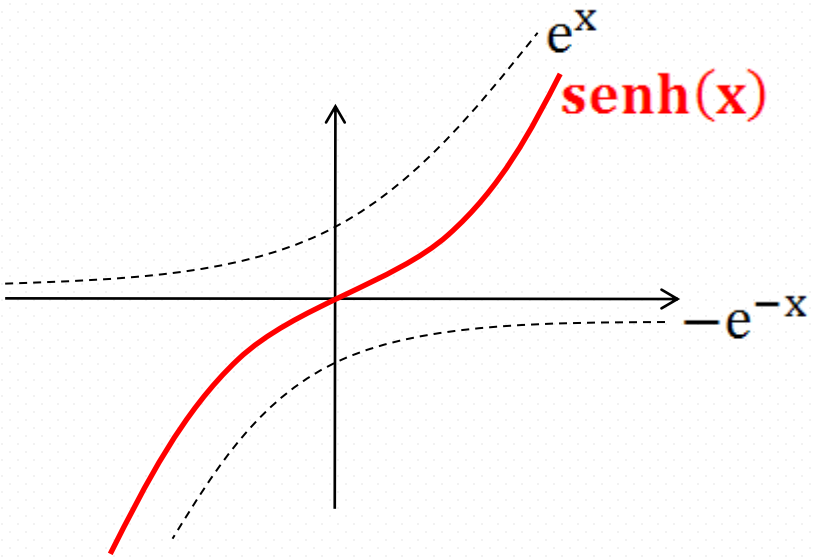


gráfico de $\mathbf{f(x) = senh(x)}$

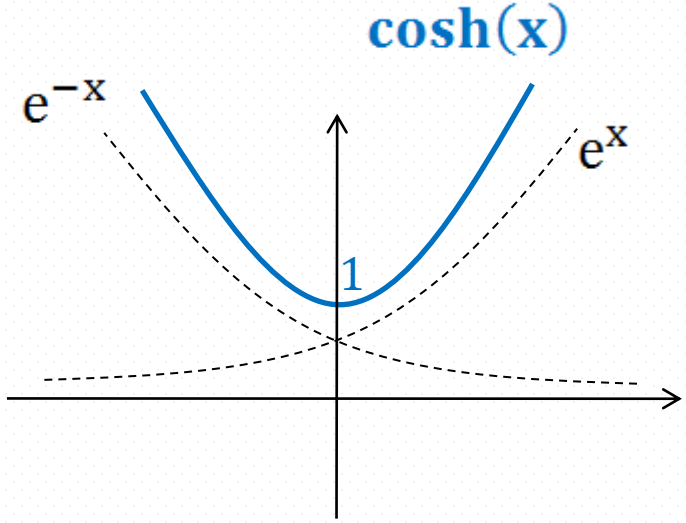


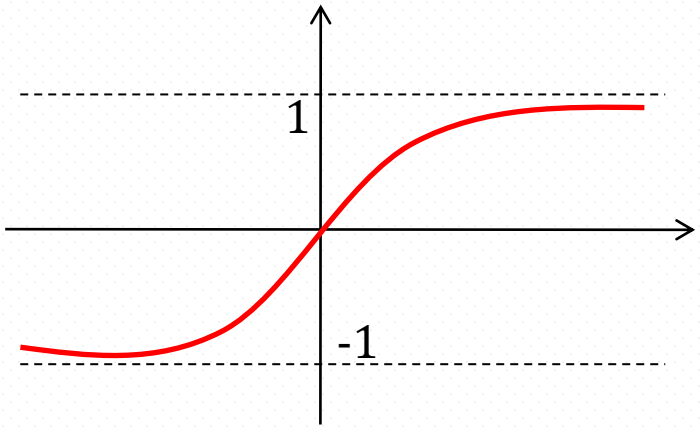
gráfico de $\mathbf{f(x) = cosh(x)}$

Outras Funções Hiperbólicas

$$\text{tgh}(x) = \frac{\text{senh}(x)}{\cosh(x)}$$

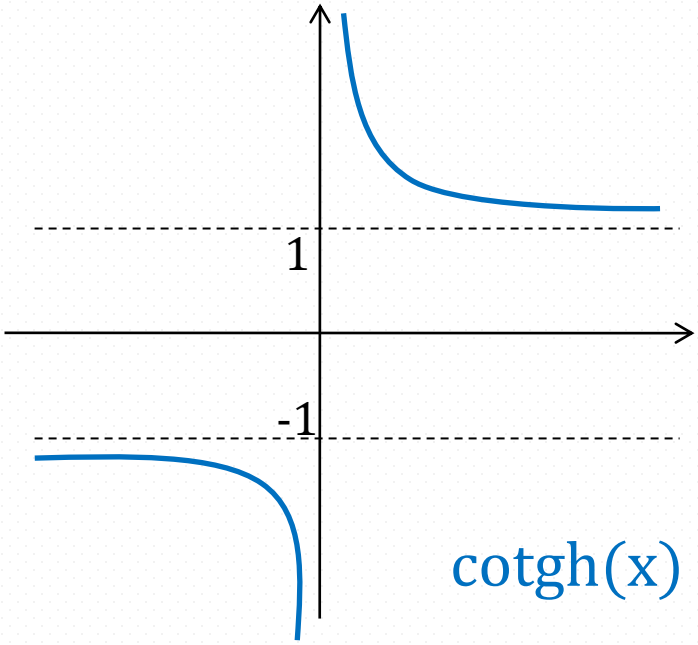
$$\text{cotgh}(x) = \frac{\cosh(x)}{\text{senh}(x)}$$

Tangente hiperbólica



tgh(x)

Cotangente hiperbólica



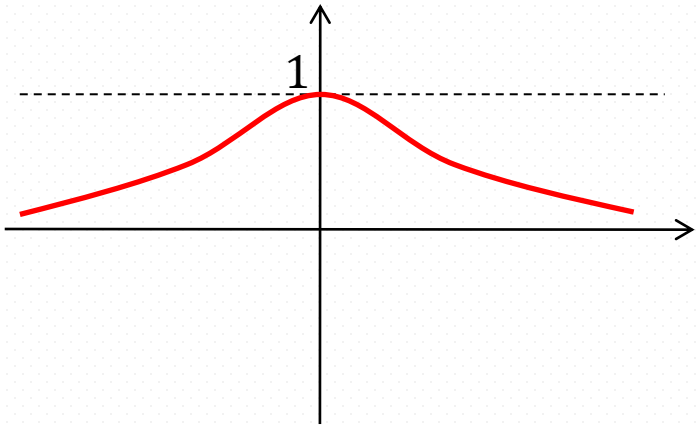
cotgh(x)

Outras Funções Hiperbólicas

$$\operatorname{sech}(x) = \frac{1}{\cosh(x)}$$

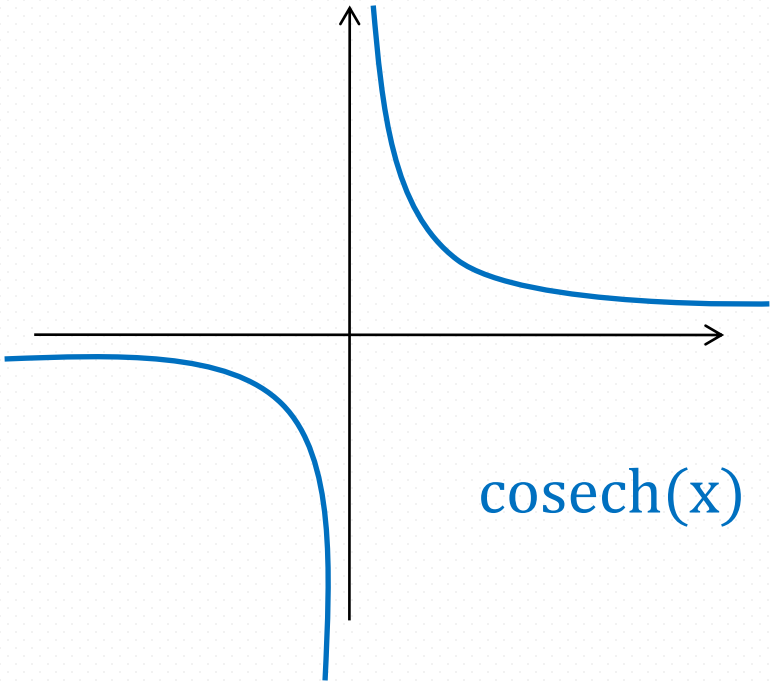
$$\operatorname{cossech}(x) = \frac{1}{\operatorname{senh}(x)}$$

Secante hiperbólica



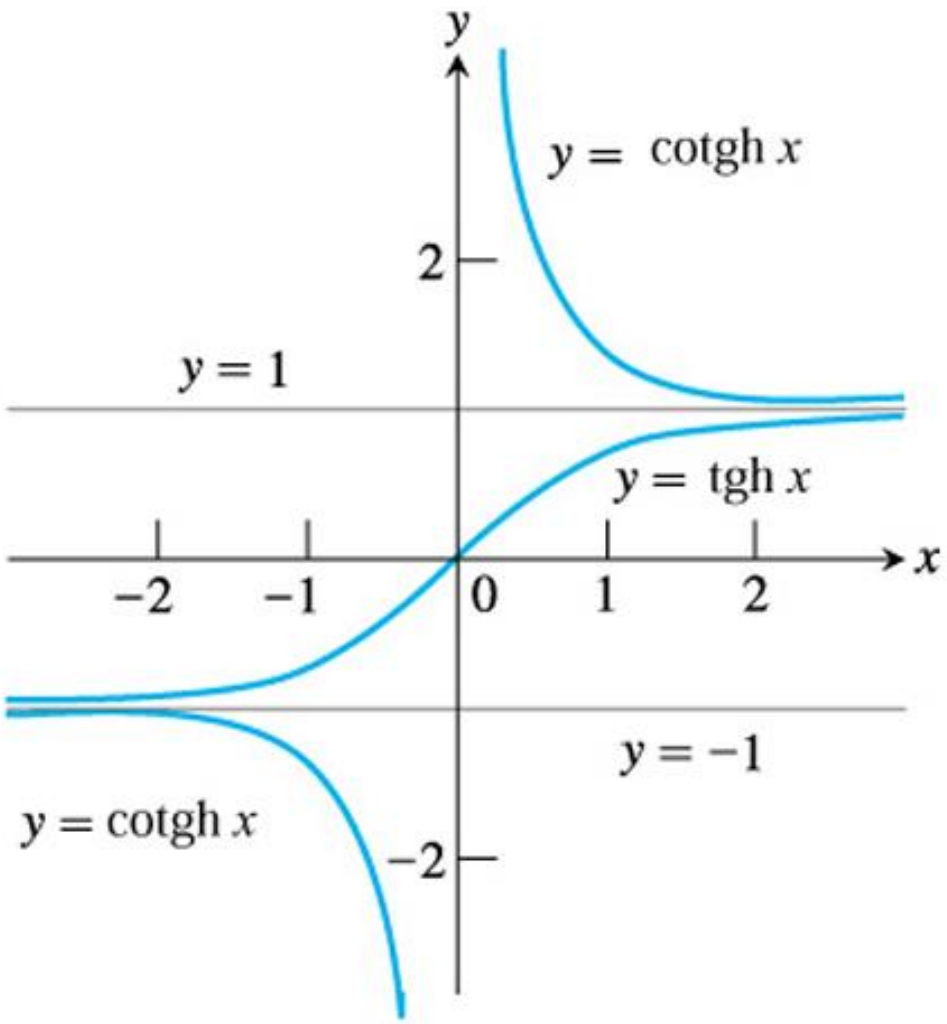
sech(x)

Cossecante hiperbólica



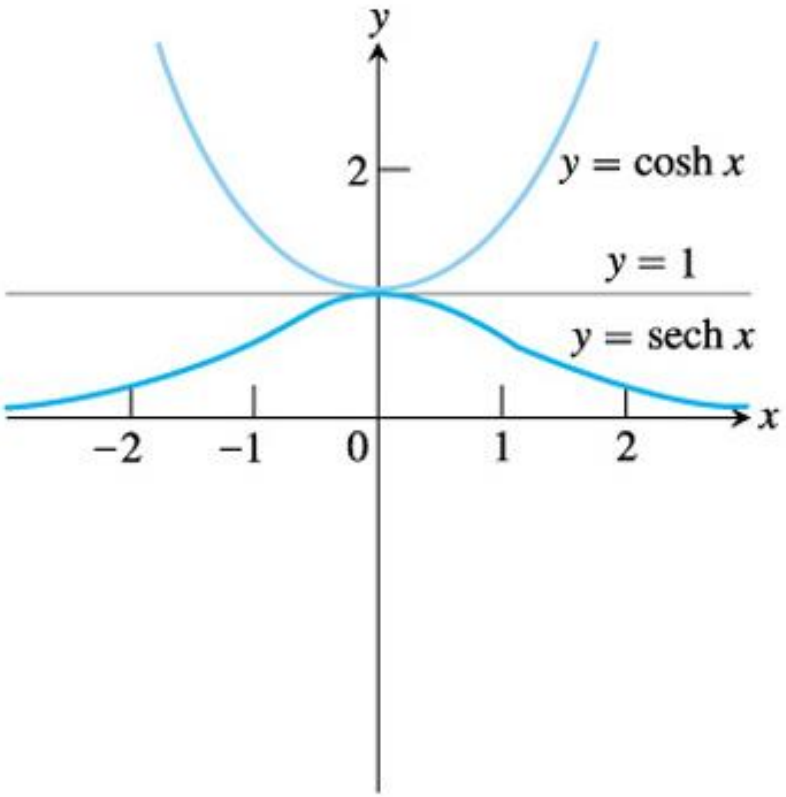
cosech(x)

Funções Hiperbólicas

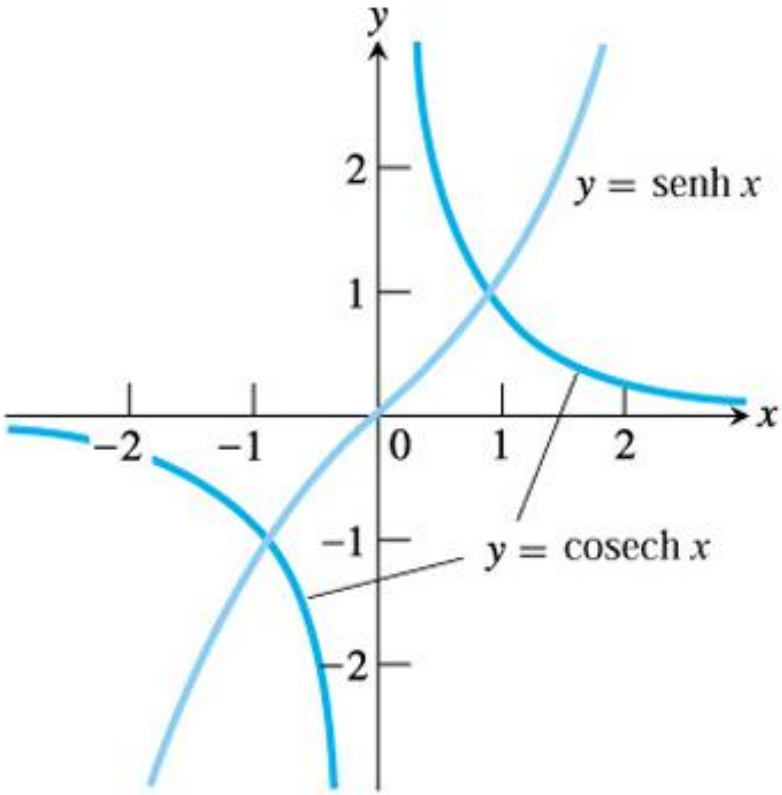


(c) Os gráficos de $y = \text{tgh}\, x$ e $y = \text{cotgh}\, x = 1/\text{tgh}\, x$

Funções Hiperbólicas



(d) Os gráficos de $y = \cosh x$ e $y = \operatorname{sech} x = 1/\cosh x$



(e) Os gráficos de $y = \operatorname{senh} x$ e $y = \operatorname{cosech} x = 1/\operatorname{senh} x$

Funções Hiperbólicas

Funções Hiperbólicas Básicas

Cosseno Hiperbólico: $\cosh x = \frac{e^x + e^{-x}}{2}$

Seno Hiperbólico: $\operatorname{senh} x = \frac{e^x - e^{-x}}{2}$

Tangente Hiperbólico: $\operatorname{tgh} x = \frac{\operatorname{senh} x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Cotangente Hiperbólico: $\operatorname{cotgh} x = \frac{\cosh x}{\operatorname{senh} x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Secante Hiperbólica: $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$

Cossecante Hiperbólica: $\operatorname{cossech} x = \frac{1}{\operatorname{senh} x} = \frac{2}{e^x - e^{-x}}$

Funções Hiperbólicas

Identidades

$$
\begin{aligned}
\operatorname{senh}(-x) &= -\operatorname{senh} x \\
\cosh(-x) &= \cosh(x) \\
\operatorname{senh} 2x &= 2 \operatorname{senh} x \cosh x \\
\cosh 2x &= \cosh^2 x + \operatorname{senh}^2 x \\
\operatorname{senh}(x+y) &= \operatorname{senh} x \cosh y + \cosh x \operatorname{senh} y \\
\cosh(x+y) &= \cosh x \cosh y + \operatorname{senh} x \operatorname{senh} y \\
\cosh^2 x &= \frac{\cosh 2x + 1}{2} \\
\operatorname{senh}^2 x &= \frac{\cosh 2x - 1}{2} \\
\cosh^2 x - \operatorname{senh}^2 x &= 1 \\
\operatorname{tgh}^2 x &= 1 - \operatorname{sech}^2 x \\
\operatorname{cotgh}^2 x &= 1 + \operatorname{cossech}^2 x
\end{aligned}
$$

Funções Hiperbólicas

Exemplo 1

Determine x sabendo que $senh(x) = 3/4$.

$$\frac{e^x - e^{-x}}{2} = \frac{3}{4}$$

$$e^x - e^{-x} = \frac{3}{2}$$

$$e^x - \frac{1}{e^x} = \frac{3}{2}$$

$$\frac{(e^x)^2 - 1}{e^x} = \frac{3}{2}$$

$$2e^{2x} - 2 = 3e^x$$

$$2y^2 - 2 = 3y$$

$$2y^2 - 3y - 2 = 0$$

$$\Delta = 9 + 16 = 25$$

$$y = \frac{3 \pm \sqrt{25}}{4}$$

$$y = \frac{3 \pm 5}{4}$$

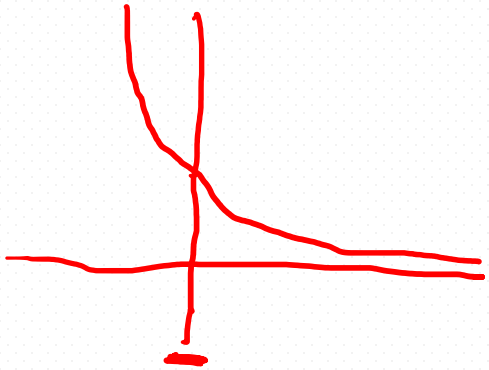
$e^x = y$

$y' = 2$

$y'' = -\frac{2}{4} = -\frac{1}{2}$

Funções Hiperbólicas

Continuação do Exemplo 1



$e^x = y$

$e^x = 2$

$\ln(e^x) = \ln(2)$

$x \cdot \ln(e) = \ln(2)$

$x = \ln(2)$ *

$e^x = -\frac{1}{2}$

$\nexists$

Funções Hiperbólicas

Exemplo 2

Resolva a equação $2\cosh(2x) + 10senh(2x) = 5$.

$$2 \cdot \left(\frac{e^{2x} + e^{-2x}}{2} \right) + 10 \cdot \left(\frac{e^{2x} - e^{-2x}}{2} \right) = 5$$

$$e^{2x} + e^{-2x} + 5e^{2x} - 5e^{-2x} = 5$$

$$6e^{2x} - 4e^{-2x} = 5$$

$$6e^{2x} - \frac{4}{e^{2x}} = 5$$

$$\frac{6e^{4x} - 4}{e^{2x}} = \frac{5e^{2x}}{e^{2x}}$$

$$6e^{4x} - 5e^{2x} - 4 = 0$$

Funções Hiperbólicas

Continuação do Exemplo 2

$6.e^{4x} - 5.e^{2x} - 4 = 0$

$6.y^2 - 5.y - 4 = 0$

$\Delta = 25 + 96 = 121$

$y = \frac{5 \pm \sqrt{121}}{12}$

$y = \frac{5 \pm 11}{12}$

$y' = \frac{4}{3}$

$y'' = -\frac{1}{2}$

$e^{2x} = y$

$e^{2x} = \frac{4}{3}$

$\ln(e^{2x}) = \ln(\frac{4}{3})$

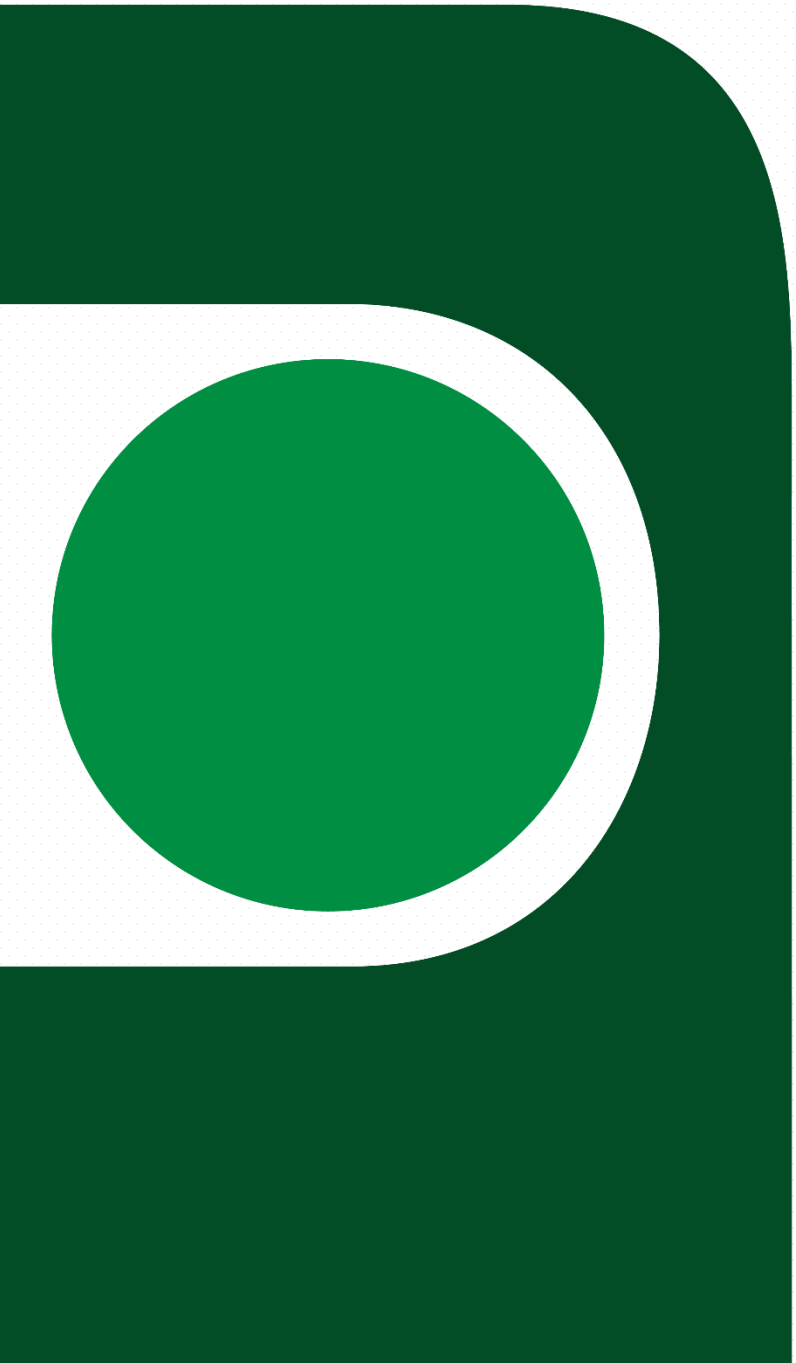
$2x = \ln(\frac{4}{3})$

$x = \frac{1}{2} \cdot \ln(\frac{4}{3})$ $= 0{,}1438$ *

$e^{2x} = -\frac{1}{2}$ ∄

Exercícios

1) Lista de Exercícios postada no Moodle.



Obrigado